

Precedence Diagram Most lines must satisfy some technological precedence requirements; that is, certain work elements must be done before the next can begin. However, most lines also allow for some latitude and more than one sequence of operations. To help you better visualize immediate predecessors, let us run through the construction of a **precedence diagram**.² We denote the work elements by circles, with the time required to perform the work shown below each circle. Arrows lead from immediate predecessors to the next work element. Example 7.4 illustrates a manufacturing process, but a back office line-flow process in a service setting can be approached similarly.

precedence diagram

A diagram that allows one to visualize immediate predecessors better; work elements are denoted by circles, with the time required to perform the work shown below each circle.

EXAMPLE 7.4 Constructing a Precedence Diagram

Green Grass, Inc., a manufacturer of lawn and garden equipment, is designing an assembly line to produce a new fertilizer spreader, the Big Broadcaster. Using the following information on the production process, construct a precedence diagram for the Big Broadcaster.

Work Element	Description	Time (sec)	Immediate Predecessor(s)
A	Bolt leg frame to hopper	40	None
B	Insert impeller shaft	30	A
C	Attach axle	50	A
D	Attach agitator	40	B
E	Attach drive wheel	6	B
F	Attach free wheel	25	C
G	Mount lower post	15	C
H	Attach controls	20	D, E
I	Mount nameplate	18	F, G
		Total 244	

SOLUTION

Figure 7.4 shows the complete diagram. We begin with work element A, which has no immediate predecessors. Next, we add elements B and C, for which element A is the only immediate predecessor. After entering time standards and arrows showing precedence, we add elements D and E, and so on. The diagram simplifies interpretation. Work element F, for example, can be done anywhere on the line after element C is completed. However, element I must await completion of elements F and G.

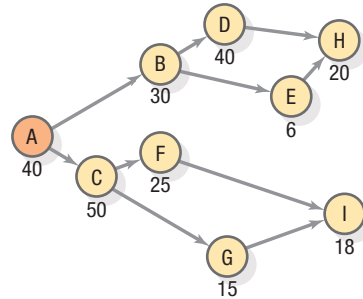


FIGURE 7.4
Precedence Diagram
for Assembling the Big
Broadcaster

DECISION POINT

Management now has enough information to develop a line-flow layout that clusters work elements to form workstations, with a goal being to balance the workloads and, in the process, minimize the number of workstations required.

Desired Output Rate The goal of line balancing is to match the output rate to the staffing or production plan. For example, if the plan calls for 4,800 units or customers per week and the line operates 80 hours per week, the desired output rate ideally would be 60 units or customers (4,800/80) per hour. Matching output to the plan ensures on-time delivery and prevents buildup of unwanted inventory or customer delays. However, managers should avoid rebalancing a line too frequently because each time a line is rebalanced many workers' jobs on the line must be re-designed, temporarily hurting productivity and sometimes even requiring a new detailed layout for some stations.

²Precedence relationships and precedence diagrams are also important in the entirely different context of project management covered in Chapter 2.

cycle time

The maximum time allowed for work on a unit at each station.

Cycle Time After determining the desired output rate for a line, the analyst can calculate the line's cycle time. A line's **cycle time** is the maximum time allowed for work on a unit at each station.³ If the time required for work elements at a station exceeds the line's cycle time, the station will be a bottleneck, preventing the line from reaching its desired output rate. The target cycle time is the reciprocal of the desired hourly output rate:

$$c = \frac{1}{r}$$

where

c = cycle time in hours per unit

r = desired output rate in units per hour

For example, if the line's desired output rate is 60 units per hour, the cycle time is $c = 1/60$ hour per unit, or 1 minute.

Theoretical Minimum To achieve the desired output rate, managers use line balancing to assign every work element to a station, making sure to satisfy all precedence requirements and to minimize the number of stations, n , formed. If each station is operated by a different worker, minimizing n also maximizes worker productivity. Perfect balance is achieved when the sum of the work-element times at each station equals the cycle time, c , and no station has any idle time. For example, if the sum of each station's work-element times is 1 minute, which is also the cycle time, the line achieves perfect balance. Although perfect balance usually is unachievable in practice, owing to the unevenness of work-element times and the inflexibility of precedence requirements, it sets a benchmark, or goal, for the smallest number of stations possible. The **theoretical minimum (TM)** for the number of stations is

$$TM = \frac{\sum t}{c}$$

where

$\sum t$ = total time required to assemble each unit (the sum of all work-element standard times)

c = cycle time

For example, if the sum of the work-element times is 15 minutes and the cycle time is 1 minute, $TM = 15/1$, or 15 stations. Any fractional values obtained for TM are rounded up because fractional stations are impossible.

Idle Time, Efficiency, and Balance Delay Minimizing n automatically ensures (1) minimal idle time, (2) maximal efficiency, and (3) minimal balance delay. Idle time is the total unproductive time for all stations in the assembly of each unit:

$$\text{Idle time} = nc - \sum t$$

where

n = number of stations

c = cycle time

$\sum t$ = total standard time required to assemble each unit

Efficiency is the ratio of productive time to total time, expressed as a percent:

$$\text{Efficiency (\%)} = \frac{\sum t}{nc}(100)$$

balance delay

The amount by which efficiency falls short of 100 percent.

Balance delay is the amount by which efficiency falls short of 100 percent:

$$\text{Balance delay (\%)} = 100 - \text{Efficiency}$$

As long as c is fixed, we can optimize all three goals by minimizing n .

³Except in the context of line balancing, *cycle time* has a different meaning. It is the elapsed time between starting and completing a job. Some researchers and practitioners prefer the term *lead time*.



Nataliya Hora/Shutterstock.com

Skoda Automotive assembly line in Mlada Boleslav. Since 1991, Skoda has been a part of Volkswagen for over 20 years.

EXAMPLE 7.5

Calculating the Cycle Time, Theoretical Minimum, and Efficiency

Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week.

- What should be the line's cycle time?
- What is the smallest number of workstations that she could hope for in designing the line for this cycle time?
- Suppose that she finds a solution that requires only five stations. What would be the line's efficiency?

SOLUTION

- First, convert the desired output rate (2,400 units per week) to an hourly rate by dividing the weekly output rate by 40 hours per week to get $r = 60$ units per hour. Then, the cycle time is

$$c = 1/r = 1/60 \text{ (hour/unit)} = 1 \text{ minute/unit} = 60 \text{ seconds/unit}$$

- Now, calculate the theoretical minimum for the number of stations by dividing the total time, Σt , by the cycle time, $c = 60$ seconds. Assuming perfect balance, we have

$$TM = \frac{\Sigma t}{c} = \frac{244 \text{ seconds}}{60 \text{ seconds}} = 4.067 \text{ or } 5 \text{ stations}$$

- Now, calculate the efficiency of a five-station solution, assuming for now that one can be found:

$$\text{Efficiency (\%)} = \frac{\Sigma t}{nc} (100) = \frac{244}{5(60)} (100) = 81.3\%$$

DECISION POINT

If the manager finds a solution with five stations that satisfies all precedence constraints, then that is the optimal solution; it has the minimum number of stations possible. However, the efficiency (sometimes called the *theoretical maximum efficiency*) will be only 81.3 percent. Perhaps the line should be operated less than 40 hours per week (thereby adjusting the cycle time) and the employees transferred to other kinds of work when the line does not operate.

MyOMLab

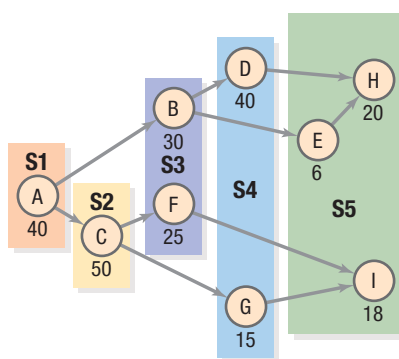
Tutor 7.1 in MyOMLab provides another example to calculate these line-balancing measures.

Finding a Solution Often, many assembly-line solutions are possible, even for such simple problems as Green Grass's. The goal is to cluster the work elements into workstations so that (1) the number of workstations required is minimized, and (2) the precedence and cycle-time requirements are not violated. The idea is to assign work elements to workstations subject to the precedence requirements so that the work content for the station is equal (or nearly so, but less than) the cycle time for the line. In this way, the number of workstations will be minimized.

Here we use the trial-and-error method to find a solution, although commercial software packages are also available. Most of these packages use different decision rules in picking which work element to assign next to a workstation being created. The ones used by POM for Windows are described in Table 7.3. The solutions can be examined for improvement, because there is no guarantee that they are optimal or even feasible. Some work elements cannot be assigned to the same station, some changes can be made to reduce the number of stations, or some shifts can provide better balance between stations.

TABLE 7.3 HEURISTIC DECISION RULES IN ASSIGNING THE NEXT WORK ELEMENT TO A WORKSTATION BEING CREATED

Create one station at a time. For the station now being created, identify the unassigned work elements that qualify for assignment: They are candidates if	
1. All of their predecessors have been assigned to this station or stations already created. 2. Adding them to the workstation being created will not create a workload that exceeds the cycle time.	
Decision Rule	Logic
<i>Longest work element</i>	Picking the candidate with the longest time to complete is an effort to fit in the most difficult elements first, leaving the ones with short times to "fill out" the station.
<i>Shortest work element</i>	This rule is the opposite of the longest work element rule because it gives preference in workstation assignments to those work elements that are quicker. It can be tried because no single rule guarantees the best solution. It might provide another solution for the planner to consider.
<i>Most followers</i>	When picking the next work element to assign to a station being created, choose the element that has the most <i>followers</i> (due to precedence requirements). In Figure 7.4, item C has three followers (F, G, and I) whereas item D has only one follower (H). This rule seeks to maintain flexibility so that good choices remain for creating the last few workstations at the end of the line.
<i>Fewest followers</i>	Picking the candidate with the fewest followers is the opposite of the most followers rule.



▲ FIGURE 7.5
Big Broadcaster Precedence
Diagram Solution

pacing

The movement of product from one station to the next as soon as the cycle time has elapsed.

Figure 7.5 shows a solution that creates just five workstations. We know that five is the minimum possible, because five is the theoretical minimum found in Example 5. All of the precedence and cycle-time requirements are also satisfied. Consequently, the solution is optimal for this problem. Each worker at each station must perform the work elements in the proper sequence. For example, workstation S5 consists of one worker who will perform work elements E, H, and I on each unit that comes along the assembly line. The processing time per unit is 44 seconds ($6 + 20 + 18$) which does not exceed the cycle time of 60 seconds (see Example 5). Furthermore, the immediate predecessors of these three work elements are assigned to this workstation or upstream workstations, so their precedence requirements are satisfied. The worker at workstation S5 can do element I at any time but will not start element H until element E is finished.

Managerial Considerations

In addition to balancing a line for a given cycle time, managers have four other considerations: (1) pacing, (2) behavioral factors, (3) number of models produced, and (4) different cycle times.

Pacing The movement of product from one station to the next as soon as the cycle time has elapsed is called **pacing**. Pacing manufacturing processes allows materials handling to be automated and requires less inventory storage area. However, it is less flexible in handling unexpected delays that require either slowing down the entire line or pulling the unfinished work off the line to be completed later.

Behavioral Factors The most controversial aspect of line-flow layouts is behavioral response. Studies show that installing production lines increases absenteeism, turnover, and grievances.

Paced production and high specialization (say, cycle times of less than 2 minutes) lower job satisfaction. Workers generally favor inventory buffers as a means of avoiding mechanical pacing. One study even showed that productivity increased on unpaced lines.

Number of Models Produced A line that produces several items belonging to the same family is called a **mixed-model line**. In contrast, a single-model line produces one model with no variations. Mixed-model production enables a plant to achieve both high-volume production *and* product variety. However, it complicates scheduling and increases the need for good communication about the specific parts to be produced at each station.

mixed-model line

A production line that produces several items belonging to the same family.

Cycle Times A line's cycle time depends on the desired output rate (or sometimes on the maximum number of workstations allowed). In turn, the maximum line efficiency varies considerably with the cycle time selected. Thus, exploring a range of cycle times makes sense. A manager might go with a particularly efficient solution even if it does not match the desired output rate. The manager can compensate for the mismatch by varying the number of hours the line operates through overtime, extending shifts, or adding shifts. Multiple lines might even be the answer.

LEARNING GOALS IN REVIEW

- 1 **Explain the theory of constraints.** Constraints or bottlenecks can exist in the form of internal resources or market demand in both manufacturing and service organizations, and in turn play an important role in determining system performance. See the section on "The Theory of Constraints (TOC)," pp. 265–267. Review opening vignette on The Radcliffe Infirmary for an application of TOC.
- 2 **Understand linkage of capacity constraints to financial performance measures.** Review and understand Table 7.1, p. 265.
- 3 **Identify bottlenecks.** The TOC provides guidelines on how to identify and manage constraints. The section "Identification and Management of Bottlenecks," pp. 267–274, shows you how to identify bottlenecks in both service as well as manufacturing firms.
- 4 **Apply theory of constraints to product mix decisions.** Review Example 7.3, pp. 270–273, to understand how using a bottleneck based method for allocating resources and determining the product mix leads to greater profits.
- 5 **Describe how to manage constraints in an assembly line.** Assembly line balancing, as a special form of a constraint in managing a line process within both manufacturing and services, can also be an effective mechanism for matching output to a plan and running such processes more efficiently. The section "Managing Constraints in a Line Process," pp. 274–279, shows you how to balance assembly lines and create work stations. Review Solved Problem 1, pp. 280–281, for an application of line balancing principles.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on processing time for average customer, bottleneck activity, maximum customers served at bottleneck per hour, average capacity of system, theoretical maximum of stations in an assembly line, longest work element decision rule, line's efficiency, and cycle time.

MyOMLab Resources	Titles	Link to the Book
Video	<i>Constraint Management at Southwest Airlines</i>	Managing Constraints Across the Organization; The Theory of Constraints
	<i>1st Bank Villa Italia: Waiting Lines</i>	Identification and Management of Bottlenecks
OM Explorer Solvers	Min-Yo Garment Company Spreadsheet	Estimate Capacity Requirements; Example 6.1 (pp. 229–230); Managing Bottlenecks in Manufacturing Processes; Example 7.2 (pp. 269–270)
OM Explorer Tutors	7.1 Line Balancing	Line Balancing; Example 7.5 (p. 277); Solved Problem 1 (pp. 280–281)
POM for Windows	Line Balancing	Line Balancing; Example 7.5 (p. 277); Solved Problem 1 (pp. 280–281)
SimQuick Simulation Exercises	Simulating process of making jewelry boxes and making choice on investing in new machine	The Theory of Constraints; Example 7.1 (pp. 267–268)
Internet Exercise	Granite Rock and Chevron	The Theory of Constraints; Identification and Management of Bottlenecks
Key Equations		
Image Library		

Key Equations

1. Cycle time: $c = \frac{1}{r}$
2. Theoretical minimum number of workstations: $TM = \frac{\sum t}{c}$
3. Idle time: $nc - \sum t$
4. Efficiency(%): $\frac{\sum t}{nc}(100)$
5. Balance delay (%): $100 - \text{Efficiency}$

Key Terms

balance delay 276
bottleneck 264
constraint 264
cycle time 276
drum-buffer-rope (DBR) 272

immediate predecessors 274
line balancing 274
mixed-model line 279
pacing 278
precedence diagram 275

theoretical minimum (TM) 276
theory of constraints (TOC) 265
throughput time 267
work elements 274

Solved Problem 1

A company is setting up an assembly line to produce 192 units per 8-hour shift. The following table identifies the work elements, times, and immediate predecessors:

Work Element	Time (Sec)	Immediate Predecessor(s)
A	40	None
B	80	A
C	30	D, E, F
D	25	B
E	20	B
F	15	B
G	120	A
H	145	G
I	130	H
J	115	C, I
	Total 720	

- What are the efficiency and balance delay of the solution found?
- Use trial and error to work out a solution, and show your solution on a precedence diagram.
- What is the theoretical minimum number of stations?
- What is the desired cycle time (in seconds)?

SOLUTION

- The sum of the work-element times is 720 seconds, so

$$TM = \frac{\sum t}{c} = \frac{720 \text{ seconds/unit}}{150 \text{ seconds/unit-station}} = 4.8 \text{ or } 5 \text{ stations}$$

which may not be achievable.

- b. Substituting in the cycle-time formula, we get

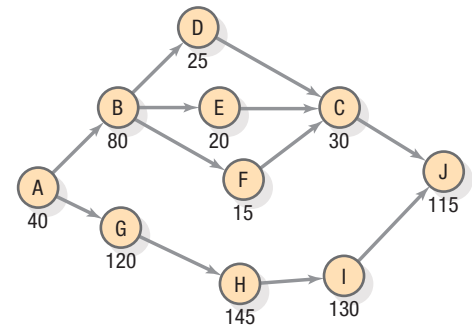
$$c = \frac{1}{r} = \frac{8 \text{ hours}}{192 \text{ units}} (3,600 \text{ seconds/hour}) = 150 \text{ seconds/unit}$$

- c. The precedence diagram is shown in Figure 7.6. Each row in the following table shows work elements assigned to each of the five workstations in the proposed solution.

- d. Calculating the efficiency, we get

$$\text{Efficiency} = \frac{\sum t}{nc} (100) = \frac{720 \text{ seconds/unit}}{5[150 \text{ seconds/unit}]} = 96\%$$

Thus, the balance delay is only 4 percent (100–96).

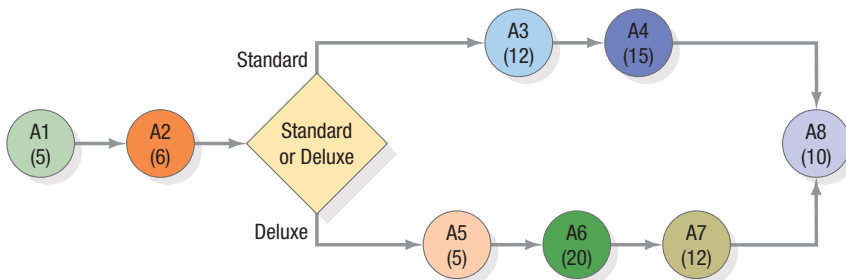


▲ FIGURE 7.6
Precedence Diagram

Station	Candidate(s)	Choice	Work-Element Time (Sec)	Cumulative Time (Sec)	Idle Time (c = 150 Sec)
S1	A	A	40	40	110
	B	B	80	120	30
	D, E, F	D	25	145	5
S2	E, F, G	G	120	120	30
	E, F	E	20	140	10
S3	F, H	H	145	145	5
S4	F, I	I	130	130	20
	F	F	15	145	5
S5	C	C	30	30	120
	J	J	115	145	5

Solved Problem 2

Bill's Car Wash offers two types of washes: Standard and Deluxe. The process flow for both types of customers is shown in the following chart. Both wash types are first processed through steps A1 and A2. The Standard wash then goes through steps A3 and A4 while the Deluxe is processed through steps A5, A6, and A7. Both offerings finish at the drying station (A8). The numbers in parentheses indicate the minutes it takes for that activity to process a customer.



- If 60 percent of the customers are Standard and 40 percent are Deluxe, what is the average capacity of the car wash in customers per hour?
- What is the capacity (measured as customers served per hour) of Bill's Car Wash to process Standard and Deluxe customers? Assume that no customers are waiting at step A1, A2, or A8.
- Which step is the bottleneck for the Standard car wash process? For the Deluxe car wash process?